

1 Quick Review

To help prepare you for your exam on Thursday, this worksheet is intended to be a review of the big ideas from Chapter 4. Below are some key ideas from the last chapter!

Linear Independence: A nonempty set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ in a vector space V is linearly independent iff

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_r\mathbf{v}_r = \mathbf{0} \quad \text{only when} \quad k_1 = k_2 = \dots = k_r = 0.$$

Span: The span of a nonempty set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ is a vector space W consisting of all possible linear combinations of S .

$$W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\} = \text{span}(S)$$

Basis: The basis for a vector space V is a set of vectors S such that

- S spans V
- S is linear independent

Dimension: Denoted by $\dim(V)$, the dimension is the number of vectors in a vector space's basis.

Row Space: Denoted by $\text{row}(\mathbf{A})$, the row space is the space spanned by the row vectors of a matrix $\mathbf{A}$.

Column Space: Denoted by $\text{col}(\mathbf{A})$, the column space is the space spanned by the column vectors of a matrix $\mathbf{A}$.

Null Space: Denoted by $\text{null}(\mathbf{A})$, the null space is the solution space of the homogenous system, $\mathbf{A}\mathbf{x} = \vec{0}$. In other contexts, it is called the *kernel*.

Rank & Nullity: The rank and nullity are the dimensions of the row space (or column space) and the null space, respectively.

Below are the **20!** equivalent statements you have learned so far!

Invertible Matrix Theorem

$\mathbf{A}$ is invertible.	$\mathbf{A}\mathbf{x} = \mathbf{0}$ has only the trivial solution.
The reduced row echelon form of $\mathbf{A}$ is I_n .	$\mathbf{A}$ is expressible as a product of elementary matrices.
$\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent for every $n \times 1$ matrix $\mathbf{b}$.	$\mathbf{A}\mathbf{x} = \mathbf{b}$ has exactly one solution for every $n \times 1$ matrix $\mathbf{b}$.
$\det(\mathbf{A}) = 0$.	The column vectors of $\mathbf{A}$ are linearly independent.
The row vectors of $\mathbf{A}$ are linearly independent.	The column vectors of $\mathbf{A}$ span $\mathbb{R}_n$.
The row vectors of $\mathbf{A}$ span $\mathbb{R}_n$.	The column vectors of $\mathbf{A}$ form a basis for $\mathbb{R}_n$.
The row vectors of $\mathbf{A}$ form a basis for $\mathbb{R}_n$.	$\mathbf{A}$ has rank n .
$\mathbf{A}$ has nullity 0.	The orthogonal complement of the null space of $\mathbf{A}$ is $\mathbb{R}_n$.
The orthogonal complement of the row space of $\mathbf{A}$ is $\{0\}$.	The kernel of T_A is $\{0\}$.
The range of T_A is $\mathbb{R}_n$.	T_A is one-to-one.

2 Problems

1. Given three points $A(0, 1, 2)$, $B(3, 5, 2)$, $C(9, 2, 0)$:

(a) Find an equation for the plane in the form $x = x_0 + t_1v_1 + t_2v_2$ that is formed by the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

(b) Find a vector $\vec{n}$ that is normal to the plane found in part (a).

(c) Find the shortest distance between the normal from part (b) and a point $P(1, 2, 3)$.

2. **(PROOF)**. Let $S = \{(x, x + y) \mid x, y \in \mathbb{R}\}$ be a set of vectors. Prove that S is a subspace of $\mathbb{R}^2$.

3. Let $\mathbf{A}$ be the matrix defined below.

$$\mathbf{A} = \begin{bmatrix} 3 & 4 & 7 \\ 1 & 2 & 4 \\ 2 & 2 & 3 \end{bmatrix}$$

(a) Find a basis for the row space, column space, and null space of $\mathbf{A}$.

(b) State the rank and nullity of $\mathbf{A}^T$.

4. Below are two different bases for $\mathbb{R}^3$.

$$A = \{(1, 2, 5), (1, 2, 0), (0, 5, 1)\}$$

$$B = \{(1, 0, 0), (0, 3, 6), (1, 5, 0)\}$$

(a) Create a basis $C \in \mathbb{P}_2$ that has the same coefficients as the basis $A \in \mathbb{R}^3$.

(b) Find the transition matrix T from basis A to B .

(c) Let $[v]_A = (1, 2, 3)$. Find $[v]_S$ and $[v]_B$.

5. Find the transformation matrix T that represents this sequence of transformations applied in $\mathbb{R}^2$:

- i) Shear up by 2.
- ii) Compression along the x-axis by a factor of 5.
- iii) Reflection about the x axis.
- iv) Rotation counterclockwise by 45° .

6. Using T from question 5, find the preimage of a vector $v = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.